SOMAIYA VIDYAVIHAR UNIVERSITY

16·05·2025 (E)

Semester: January 2025 –April 2025		
Maximum Marks: 100 **Examination: ESE Examination** **Duration:3 Hrs.**		
Programme code: 66 **Programme: Honour Programme in Artificial Intelligence**	**Class: TY**	**Semester: VI (SVU 2020)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** IT	
Course Code: 116h66C601	**Name of the Course: Deep Learning**	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory **3) Assume suitable data wherever necessary**		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	State and explain two challenges in traditional machine learning that motivate the adoption of deep learning approaches.	5
ii)	What is the intuition behind stochastic gradient descent (SGD). Why is it preferred over batch gradient descent for large datasets?	5
iii)	Differentiate between population and sample in the context of machine learning. Why is this distinction important?	5
iv)	What is selection bias in machine learning? How does it affect model performance?	5
v)	Explain the role of conditional probability in probabilistic machine learning models.	5
vi)	List and briefly explain any two resampling methods used in machine learning.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	What is backpropagation? How does it relate to gradient-based learning?	5
ii)	What are vanishing gradients? Why is it a challenge in deep networks?	5
	OR	
Q2 A	What is XOR problem? Elaborate on how a deep feedforward neural network can solve it with a simple network architecture.	10
Q 2 B	Solve any **One**	**10**
i)	What are the key challenges in optimizing deep neural networks? What are the different strategies used for optimization?	10
ii)	Explain the idea behind Tangent Distance, Tangent Propagation, and Manifold Tangent Classifier. How do these methods enhance learning in neural networks?	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	A convolutional neural network has the following layer configuration: • Input: 64×64×3 • Conv Layer: 32 filters of size 3×3, stride = 1, padding = 1 • Max Pooling: 2×2, stride = 2 • Conv Layer: 64 filters of size 3×3, stride = 1, padding = 1 • Max Pooling: 2×2, stride = 2 Calculate the size of the output feature map after each layer.	10
ii)	Given a 6×6 input and a 3×3 kernel (no padding, stride = 1), apply the convolution operation step-by-step to compute the output feature map. Use the following input matrix and kernel: Input matrix: [[1, 2, 3, 0, 1, 2], [0, 1, 2, 3, 1, 0], [1, 0, 1, 2, 0, 1], [2, 1, 0, 1, 2, 3], [0, 1, 2, 1, 0, 1], [1, 2, 3, 0, 1, 2]] Kernel: [[1, 0, -1], [1, 0, -1], [1, 0, -1]]	10
iii)	A convolutional neural network is given with the following configuration: • Conv Layer 1: 32 filters, 3×3, stride = 1, input = 64×64×3 • Conv Layer 2: 64 filters, 3×3, stride = 1, no padding • Fully Connected Layer: output size = 128, input from a flattened 16×16×64 volume Calculate: a) Output size after each convolutional layer b) Total number of trainable parameters in the full network	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Elaborate in detail the structure and working of an LSTM network. How does it solve the problem of vanishing gradients?	10
ii)	Explain the Encoder-Decoder architecture used in sequence-to-sequence tasks. How is it helpful in applications like machine translation?	10
iii)	Describe the working of a Recursive Neural Network and highlight its application in Natural Language Processing.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	20
i)	Describe likelihood in the context of machine learning models.	5
ii)	List any two variants of the basic convolution function and mention one use case for each.	5
iii)	Explain the concept of unfolding a computational graph in the context of Recurrent Neural Networks (RNNs).	5
iv)	Differentiate between max pooling and average pooling with one advantage of each.	5
v)	How does early stopping help in preventing overfitting in deep models?	5
vi)	Differentiate between Recurrent Neural Networks (RNNs) and Recursive Neural Networks.	5